ProPath — Project Proposal & System Design (Milestone Document)

Career preparation & professional development

Course milestone — combined team document
Team name: BlackCS
Document version: 2.0 · **Date:** March 25, 2026
GitHub repository: https://github.com/Wisesofthemall/FocusFlow

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1. Project Overview

1.1 Problem statement

Job seekers and professionals building their careers often juggle **applications, interviews, follow-ups, and skill goals** across spreadsheets, email, bookmarks, and memory. That **fragmentation** leads to **missed deadlines**, **dropped opportunities**, and **stress**. The problem is worth solving because clearer visibility and prioritization improve outcomes without forcing users onto heavyweight enterprise tools.

1.2 Target users

- **Primary:** Students and early-career professionals preparing to enter or change fields.
- **Secondary:** Anyone organizing job applications, networking, and professional growth alongside other commitments.

1.3 Value proposition

ProPath gives users **one dashboard** where **job applications** are **organized**, **editable**, and **automatically ordered by urgency** so "what needs attention today" is obvious. Connecting **Google Calendar** surfaces **real schedule context** (interviews, career fairs, reminders, synced events) so users can **plan next actions around fixed commitments** instead of guessing.

1.4 Minimum viable product (MVP)

Area	MVP capability
Auth	User signup and login (secure sessions; password or OAuth per course policy).
Applications	Create, edit, delete **job applications**; fields include **company**, **role/title**, **status** (e.g. applied / interviewing / offer / rejected), **next action or deadline** for sorting and reminders.
Priority / sorting	**Auto-sort** by urgency (e.g. next action or deadline ascending, with optional priority field or overdue boost).
Dashboard	**Today / this week** view: applications needing attention soon (and optionally overdue follow-ups), sorted.
Calendar integration	User **connects** Google Calendar (OAuth); app **fetches upcoming events** and shows them alongside applications (read-only for MVP is acceptable).

This MVP is the **smallest** version that still solves the core problem: clarity + today-focused execution + schedule awareness for career preparation.

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2. External API integration (required)

2.1 Planned API: Google Calendar API

- **Data provided:** Calendar **events** (title, start/end times, optional location/description), for the user's selected calendar(s).
- **How ProPath uses it (meaningful to the product):**
  - After the user authorizes access, the backend retrieves **upcoming events** for a configurable window (e.g. next 7–14 days).
  - The **dashboard** displays **events near "today"** next to **upcoming application actions**, so users see **interviews, networking blocks, and fixed commitments** when deciding what to do first.
  - **Applications** remain the source of truth for the job search; events provide **schedule context** (not unrelated decoration).

2.2 Fallback option

If OAuth scope or timeline is too heavy for the semester, the team may use a **public holidays API** (country/region) to flag days that disrupt routines. That remains schedule-relevant but is weaker than per-user calendar; the proposal would be updated to state this choice explicitly.

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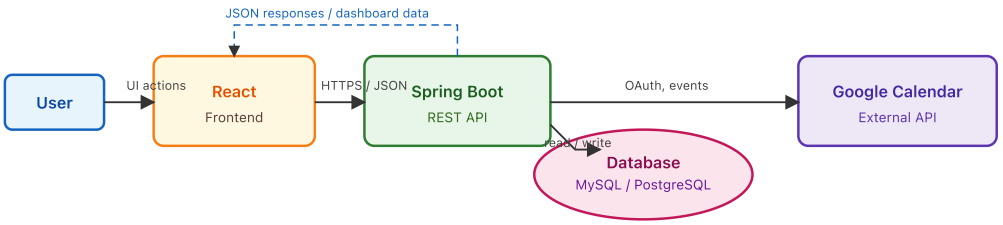
3. System design

3.1 System modules (major components)

1. **Authentication module** — Registration, login, session/JWT issuance, optional OAuth for Google.
2. **Application tracker module** — CRUD for job applications; validation; maps to persistence layer.
3. **Calendar integration module** — OAuth token storage (secure), sync job or on-demand fetch from Google Calendar API, normalization to internal "event" representation.
4. **Dashboard module (API + UI)** — Aggregates **applications due soon** (and optional "this week") plus **near-term calendar events** for the logged-in user.
5. **Sorting / priority service** — Central rules for urgency (next action or deadline, overdue, optional explicit priority) so list and dashboard stay consistent.

3.2 Level-0 data flow diagram (DFD)

The diagram below shows **user**, **React frontend**, **Spring Boot API**, **database**, and **external API**.



Narration: The user interacts with the React app. The frontend sends authenticated requests to the Spring Boot API for applications and dashboard data. The API persists users and applications in the database and, when authorized, calls the Google Calendar API to retrieve events, merges that data for the dashboard, and returns JSON to the client.

Mermaid source (optional, for tools that render Mermaid):

```
flowchart LR
  subgraph actors [Actors]
    User[User]
  end
  subgraph client [Client]
    React[React_Frontend]
  end
  subgraph server [Server]
    API[Spring_Boot_API]
  end
  subgraph data [Data stores]
    DB[(Database)]
  end
  subgraph external [External]
    GCal[Google_Calendar_API]
  end
  User -->|"applications_auth_actions"| React
  React -->|"HTTPS_JSON"| API
  API -->|"read_write"| DB
  API -->|"OAuth_token_events"| GCal
  API -->|"responses"| React
  React -->|"UI_updates"| User
```

3.3 Entities / data objects and relationships

User

Attribute	Description
`id`	Primary key (UUID or long).
`name`	Display name.
`email`	Unique login identifier.
`passwordHash`	If using password auth (never store plain text).
`googleRefreshToken`	Encrypted/stored only if using Calendar OAuth (optional field).

Application (job application / opportunity)

Attribute	Description
`id`	Primary key.
`userId`	Foreign key → User.
`company`	Employer or organization name.
`roleTitle`	Job title or role applied for.

`status`	e.g. APPLIED / INTERVIEWING / OFFER / REJECTED / WITHDRAWN (team-defined enum).
`nextActionDate`	Date/time for follow-up, deadline, or interview; drives sorting and “today” filter.
`priority`	e.g. LOW / MEDIUM / HIGH (optional for MVP).
`createdAt` / `updatedAt`	Audit fields.

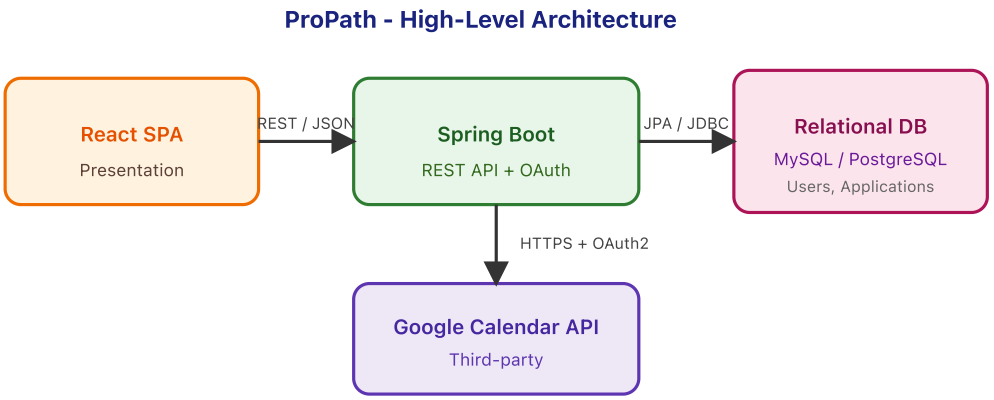
CalendarEvent (DTO or cached entity; full table optional in MVP)

Attribute	Description
`id`	External id from Google or composite key.
`userId`	Owner.
`title`	Event summary.
`start` / `end`	Instant or datetime.
`source`	e.g. `GOOGLE`

Relationships

- **User 1 — N Application:** Each application belongs to one user.
- **User 1 — N CalendarEvent (logical):** Events belong to the user; MVP may fetch live from Google and map to DTOs without persisting, or cache rows for performance—the team will document the chosen approach in the implementation milestone.

3.4 Architecture diagram



Caption: Single-page React client communicates only with the Spring Boot API; the API owns database access and external calendar calls.

Mermaid source (optional):

```
flowchart TB
  subgraph presentation [Presentation]
    ReactApp[React_SPA]
  end
  subgraph application [Application]
    SpringBoot[Spring_Boot_REST_API]
  end
  subgraph persistence [Persistence]
    RDBMS[(MySQL_or_PostgreSQL)]
  end
  subgraph thirdParty [Third_Party]
    GoogleCal[Google_Calendar_API]
  end
  ReactApp -->|REST_JSON_HTTPS| SpringBoot
  SpringBoot -->|JDBC_JPA| RDBMS
  SpringBoot -->|HTTPS_OAuth2| GoogleCal
```

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4. Demo-centric planning

Demo storyboard (aligns with rubric)

1. **Introduce the problem** — Fragmented job search data and weak “what needs attention today” visibility.
2. **Show the solution** — ProPath dashboard combining **applications** and **calendar** context.
3. **Live flow**
    - Log in (or sign up, then log in).
    - Add two or three **applications** with different **next actions or deadlines** (one due today, one tomorrow, one next week).
    - Show **auto-sort** (list or dashboard reordering by urgency).
    - **Connect Google Calendar** (or use a pre-connected demo account); **upcoming events** appear.
    - Return to the dashboard: **Today’s priorities** plus **today’s / near-term events** are visible together.
    - Update an application **status** (e.g. to interviewing or done); show **UI refresh** and **persisted** state after reload or re-fetch.

The MVP supports **user → creates record → system updates DB → UI shows sorted/dashboard view** for evaluators.

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5. Responsible AI usage

5.1 Integrity statement

BlackCS used **ChatGPT** to brainstorm **app ideas** and to **refine** documentation after the course assigned the **Career Preparation & Professional Development** theme. The team **evaluated** suggestions, chose **ProPath** (branded name for this product), and **wrote and reviewed** the proposal, diagrams, and wireframes. API choice, entities, and architecture are **BlackCS decisions** after critical review of AI output.

5.2 AI usage log (appendix)

Current log for this milestone; add rows below if the team uses additional AI tools in later work.

Date	Tool	Prompt summary (paraphrased)	Purpose	How results influenced BlackCS design
March 24, 2026	ChatGPT	Asked for app ideas that would satisfy the full-stack milestone (real problem, MVP, external API, system design, demo-friendly flow, feasible for a semester)	Ideation and early scoping	Early direction toward a planner with calendar integration; later **pivoted** to career prep per course assignment.
March 25, 2026	ChatGPT	Asked to refactor milestone docs for **Career Preparation & Professional Development**, product name **ProPath**, team roles and entities	Documentation refresh after project theme change	Helped structure **Application** entity, module names, and demo storyboard consistent with **React + Spring Boot + DB + Google Calendar API**.

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6. Design artifacts (wireframes)

Low-fidelity wireframes for core screens. Reference as **Figure 1–4** in presentations or written reports.

Figure	File	Description
1	wireframes/01-login.svg	Login
2	wireframes/02-tasks.svg	Application list with add/edit/delete
3	wireframes/03-dashboard-today.svg	Dashboard — Today (career actions + calendar context)
4	wireframes/04-calendar-connect-events.svg	Connect Google Calendar + upcoming events

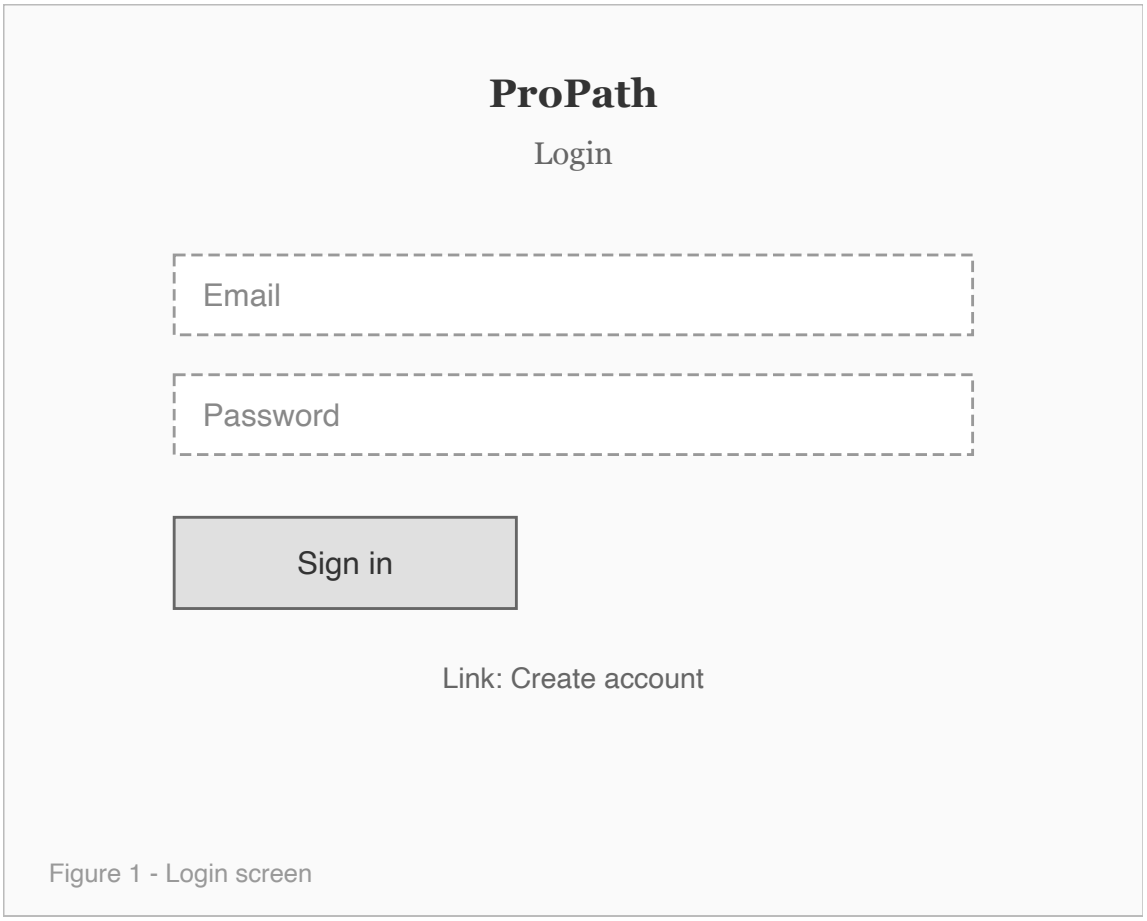

Figure 1 - Login screen

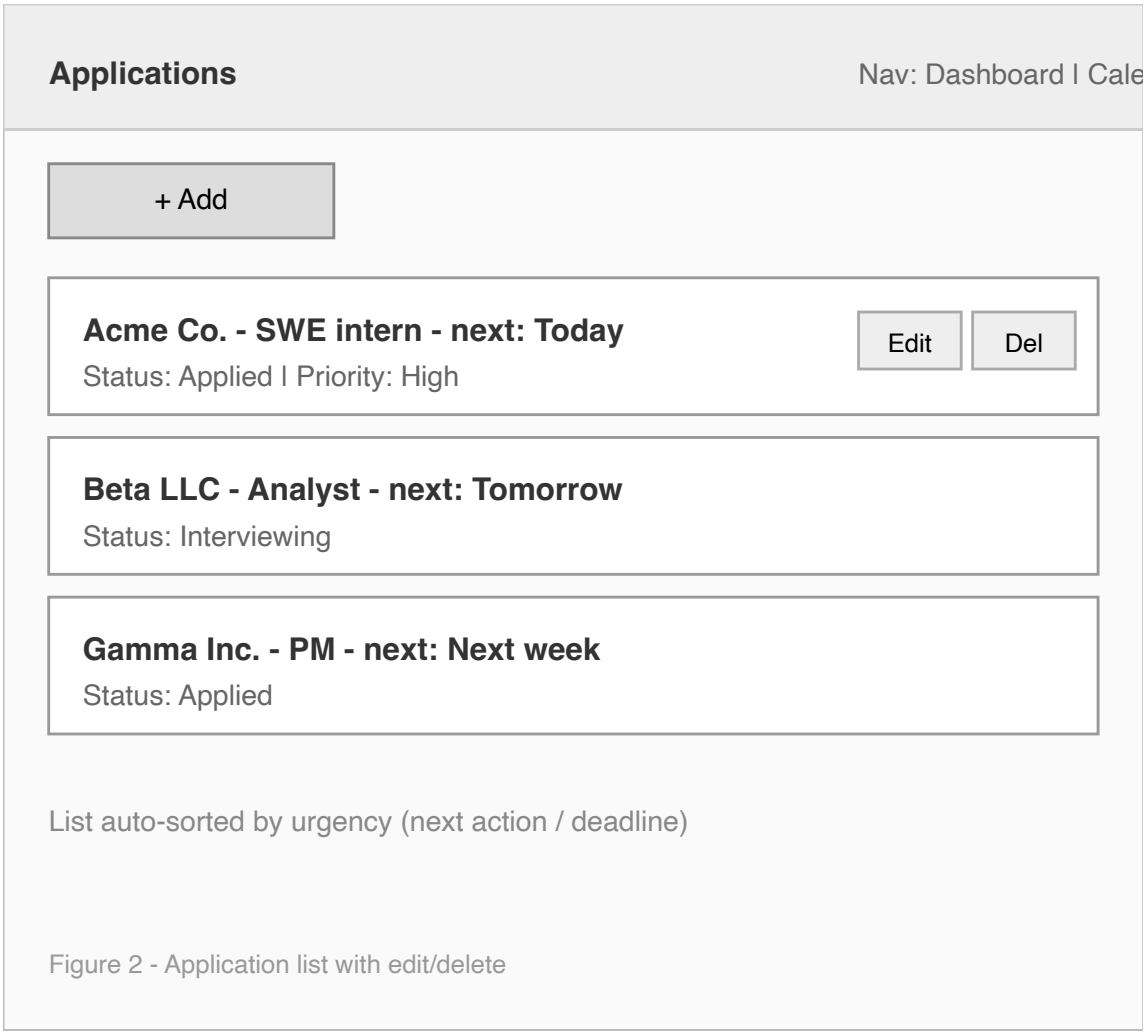

Figure 2 - Application list with edit/delete

Today Nav: Applications I Calenda

Today's priorities

Acme Co. - SWE intern

Next action today I [ ] Done

Overdue: Follow up - Beta LLC (yesterday)

Upcoming (calendar)

2:00 PM - Phone screen (from calendar)

4:30 PM - Career fair networking

Merged view: next actions + near-term events

Figure 3 - Dashboard (Today)

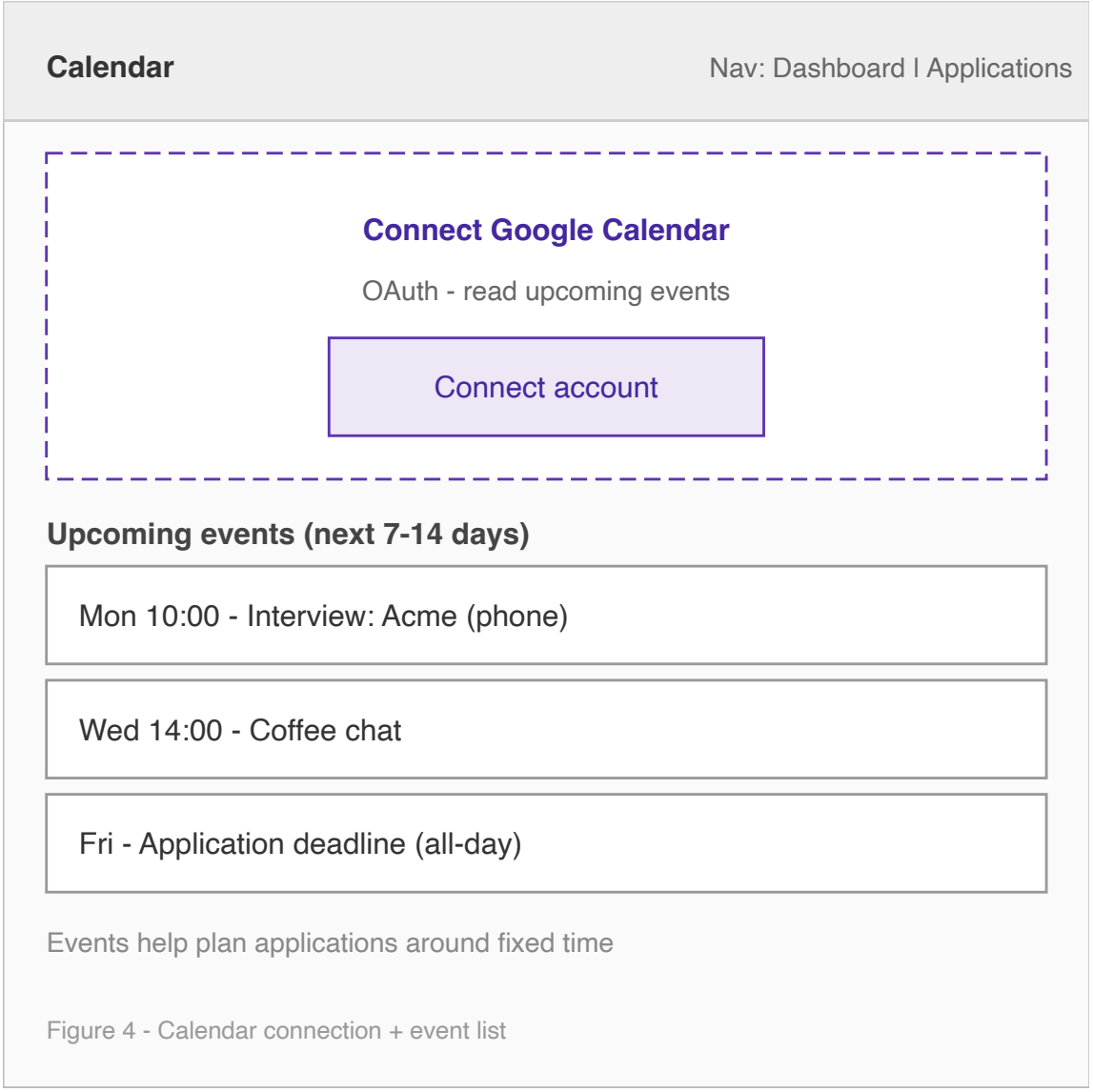

Figure 4 - Calendar connection + event list

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7. Project planning

Team: BlackCS

7.1 Team member roles

Member name (BlackCS)	Responsibility
Lovinson Dieujuste	Frontend (React), UI states, API client, demo polish
Malcolm Richards	Backend (Spring Boot), authentication, application REST APIs
Terrance Holloway	Database schema, migrations, JPA entities; Google Calendar OAuth and integration; security review; diagrams and wireframes

7.2 Collaboration & repository

- **Team:** BlackCS
- **GitHub:** https://github.com/Wisesofthemall/FocusFlow
- **Practices:** feature branches, pull requests, README updates per milestone, shared doc review before due dates.

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8. Success criteria checklist (self-review)

- [ ] Problem statement explains real pain and why it matters
- [ ] Full-stack design is feasible: **React + Spring Boot + MySQL/PostgreSQL + Google Calendar API**
- [ ] System modules, **Level-0 DFD**, **entities** with **relationships**, **architecture** diagram included
- [ ] External API **meaningfully** supports planning around the user's schedule (interviews, events, deadlines)
- [ ] Demo path is clear from MVP features
- [ ] **AI usage log** completed accurately
- [ ] **Team name (BlackCS)** and **individual names** in Section 7, **repo link** correct, **wireframes** attached or embedded

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9. Repository artifacts (this project)

Artifact	Path
Milestone (source)	ProPath-Milestone-Document.md
Milestone (PDF)	ProPath-Milestone-Document.pdf
Wireframes (SVG)	wireframes/
Diagrams (SVG)	diagrams/